1 Introduction

Many applications face the situation where a set $S = \{X_1, \dots, X_d\}$ of variables has to be divided into clusters in such a way that inside each cluster the variables are dependent, but the clusters between theselves are as independent as possible. For example, in environmental sciences, $\mathbf{X}_j$ can be a series of recording of precipitation, temperature or wind speed. The goal is to find the regions such that different regions evolve independently, but within each region the quantities of interest are mutually informative.

(à modifier) The problem of partitioning a random vector $\mathbf{X} = (X_1, \dots, X_d) \in \mathbb{R}^d$ in groups of variable that are similar in nature is known as variable clustering. Solutions to this problem are typically algorithmic in nature. Given $\mathbf{X}_1, \dots, \mathbf{X}_n$ observations on $\mathbf{X}$, one defines a dissimilarity measure between the data vectors of observations on two components X_j and X_l of $\mathbf{X}$, and places the two variables in the same cluster if the data dissimilarity measure is small. The most popular measure is the Euclidean distance between vectors in $\mathbb{R}^n$ or, equivalently, the negative sample correlation and is typically the measure employed in popular clustering algorithms such as hierarchical clustering ([Saunders et al., 2021]) or K-means ([Janßen and Wan, 2020]) and their variants ([Bernard et al., 2013, Bador et al., 2015]).

In this work, we have a different point of view, model based clustering for independence clustering. The problem of independence clustering was explicitly stated in [Ryabko, 2017] is the following :

Given a set $S = \{X_1, \dots X_d\}$ of random variables, it is required to find the finest partitioning $\{\boldsymbol{X}^{(1)}, \dots, \boldsymbol{X}^{(K)}\}$ of S into clusters such that the clusters $\{\boldsymbol{X}^{(1)}, \dots, \boldsymbol{X}^{(K)}\}$ are mutually independent.

2 A model for variable clustering

Let $\mathbf{X} \in \mathbb{R}^d$ be an extreme value random vector. Let $O = \{O_k\}_{k=1,\dots,K}$ be a partition of $\{1, \dots, d\}$, for some integer K, $1 \leq K \leq d$. In what follows, we introduce extreme value theory and a model for variable clustering.

2.1- extreme value theory A d-dimensional random vector $\mathbf{X} = (X_1, \dots, X_d)$ follows the law of a multivariate extreme-value distribution if its one dimensional marginal distributions $G_j(x) = \mathbb{P}\{X_j \leq x\}$ for all $x \in \mathbb{R}$ and $j \in \{1, \dots, d\}$ are Generalized extreme value distributions, and the joint distribution can be written, for all $\mathbf{x} \in \mathbb{R}^d$, in the form

$$G(\mathbf{x}) = C\left(G_1(x_1), \dots, G_d(x_d)\right), \tag{1}$$

where C is an extreme value copula, *i.e.*, for all $\mathbf{u} \in (0, 1]^d$

$$C(\mathbf{u}) = \exp\left\{-L\left(-\ln(u_1), \dots, -\ln(u_d)\right)\right\},$$

with L is known as the stable tail dependence function (see [Gudendorf and Segers, 2010] for an overview of extreme value copulae). As it is a homogeneous function of order 1, *i.e.* $L(a\mathbf{z}) = aL(\mathbf{z})$ for all $a > 0$, we have, for all $z \in [0, \infty)^d$,

$$L(\mathbf{z}) = (z_1 + \cdots + z_d)A(\mathbf{t}),$$

with $t_j = z_j/(z_1 + \ldots z_d)$ for $j \in \{2, \ldots, d\}$ and $t_1 = 1 - t_2 - \cdots - t_d$, and A is the restriction of L into the d-dimensional unit simplex, viz.

$$S_d = \{(v_1, \ldots, v_d) \in [0, 1]^d : v_1 + \cdots + v_d = 1\}.$$

The function A is known as the Pickands dependence function and is often used to quantitfy the extremal dependence among the element of X. Indeed, A satisfies the constraints $1/d \leq \max(t_1, \ldots, t_d) \leq A(\mathbf{t}) \leq 1$ for all $\mathbf{t} \in S_d$, with lower and upper bounds corresponding to the complete dependence and independence cases.

2.2- AI-block models Motivated by a rich set of applications, we consider variable clustering as the initial dimension reduction step applied to the observed vector $\mathbf{X} = (X_1, \ldots, X_d)$. These models are build on the assumption that the observed variables $\mathbf{X} = (X_1, \ldots, X_d)$ can be partionned into K unknown clusters $O = \{O_1, \ldots, O_K\}$ such that variables in the same cluster are as dependent as possible and the clusters are mutually independent. We define a population-level cluster as a group of variables that shares the same dependence extremal structure within the cluster and is independent from the other clusters.

To keep the presentation focused, let O be a partition of $\{1, \ldots, d\}$ into K groups and let $k : \{1, \ldots, d\} \to \{1, \ldots, K\}$ be an index assignement function defined by $O_k = \{a : k(a) = k\} = \{i_{k,1}, \ldots, i_{k,d_k}\}$ with $d_1 + \cdots + d_K = d$. Consider $\mathbf{X}^{(1)}, \ldots, \mathbf{X}^{(K)}$ be extreme value random vectors with extreme value copulae $C^{(1)}, \ldots, C^{(K)}$ respectively. We suppose that $\mathbf{X}^{(k)}$ and $\mathbf{X}^{(j)}$ are mutually independent if $k \neq j$. Let us define the following function :

$$\begin{array}{rccl} C: & [0,1]^d & \longrightarrow & [0,1] \\ & \mathbf{u} & \longmapsto & \Pi_{k=1}^{K} C^{(k)}(u_{i_{k,1}}, \ldots, u_{i_{k,d_k}}). \end{array}$$

We want to show that C is an extreme value copula associated to $\mathbf{X} = (\mathbf{X}^{(1)}, \ldots, \mathbf{X}^{(K)})$ in order that all objects we use below are well-defined, in particular the existence of a *stable tail dependence function* associated to C.

Lemma 1. *C is an extreme value copula associated to the random vector $\boldsymbol{X}$.*

Proof We first show that C is a copula function. It is clear that $C(\mathbf{u}) \in [0, 1]$ for every $\mathbf{u} \in [0, 1]^d$. We check that its univariate marginals are uniformly distributed on $[0, 1]$. Without loss of generaly,

take $u_{i_{1,1}} \in [0,1]$ and let us compute

$$C(1,\dots,u_{i_{1,1}},\dots,1) = C^{(1)}(u_{i_{1,1}},1,\dots,1)\Pi_{k=1}^{K} C^{(k)}(1,\dots,1) = C^{(1)}(u_{i_{1,1}},1,\dots,1) = u_{i_{1,1}}.$$

So C is a copula function. We now have to prove that C is an extreme value copula. We recall that C is an extreme value copula if and only if C is max-stable, that is for every $m \geq 1$

$$C(u_1,\dots,u_d) = C(u_1^{1/m},\dots,u_d^{1/m})^m.$$

By definition, we have

$$C(u_1^{1/m},\dots,u_d^{1/m})^m = \left(\Pi_{k=1}^{K} C^{(k)}\left(u_{i_{k,1}}^{1/m},\dots,u_{i_{k,d_k}}^{1/m}\right)\right)^m = \Pi_{k=1}^{K}\left\{C^{(k)}\left(u_{i_{k,1}}^{1/m},\dots,u_{i_{k,d_k}}^{1/m}\right)\right\}^m.$$

Using that $C^{(1)},\dots,C^{(K)}$ are extreme value copulae, thus max stable, we obtain

$$C(u_1^{1/m},\dots,u_d^{1/m})^m = \Pi_{k=1}^{K} C^{(k)}\left(u_{i_{k,1}},\dots,u_{i_{k,d_k}}\right) = C(u_1,\dots,u_d).$$

Thus C is an extreme value copula. We end the proof by proving that C is associated to the random vector $\mathbf{X} = (\mathbf{X}^{(1)},\dots,\mathbf{X}^{(K)})$, that is

$$\mathbb{P}\left\{\mathbf{X} \leq \mathbf{x}\right\} = C(G_1(x_1),\dots,G_d(x_d)), \quad \mathbf{x} \in \mathbb{R}^d.$$

Using mutual independence between random vectors, we have

$$\begin{aligned}
\mathbb{P}\left\{\mathbf{X} \leq \mathbf{x}\right\} &= \Pi_{k=1}^{K}\mathbb{P}\left\{X_{i_{k,1}} \leq x_{i_{k,1}},\dots,X_{i_{k,d_k}} \leq x_{i_{k,d_k}}\right\} \\
&= \Pi_{k=1}^{K} C^{(k)}\left(G_{i_{k,1}}(x_{i_{k,1}}),\dots,X_{i_{k,d_k}} \leq x_{i_{k,d_k}}\right) \\
&= C(G_1(x_1),\dots,G_d(x_d)).
\end{aligned}$$

Hence the result. □

As C is an extreme value copula by Lemma 1, there exists a *stable tail dependence function* L say which is associated with it. One can show that L can be expressed as a sum of the others *stable tail dependence function* using the mutual independency :

$$L\left(z_1,\dots,z_d\right) = \sum_{k=1}^{K} L^{(k)}\left(z_{i_{k,1}},\dots,z_{i_{k,d_k}}\right), \quad \mathbf{z} \in [0,\infty)^d, \tag{2}$$

where $L^{(1)},\dots,L^{(K)}$ are the stable tail dependence function associated to the extreme value copulae $C^{(1)},\dots,C^{(K)}$ respectively. Furtermore, this model is a particular form of the nested extreme value copula and is the object of the remark below.

Remark 1. *Equation (2) can restated as*

$$L(\boldsymbol{z}) = L^{(0)}\left(L^{(1)}\left(z_{i_{1,1}}, \ldots, z_{i_{1,d_1}}\right), \ldots, L^{(K)}\left(z_{i_{K,1}}, \ldots, z_{i_{K,d_K}}\right)\right),$$

where $L^{(0)}(z_1, \ldots, z_K) = \sum_{k=1}^{K} z_k$ *is a* stable tail dependence function *corresponding to the asymptotic independence. Since* C *is an extreme value copula by Lemma 1, we obtain that* C *is also a nested extreme value copula as formulated in [Hofert et al., 2018].*

We introduce the following partial order on sets, let $O = \{O_k\}_k$, $S = \{S_k\}_k$ be two partitions of $\{1, \ldots, d\}$. We say that S is a sub-partition of O if for each k' there exists k such that $S_{k'} \subseteq O_k$. We define the partial order $\leq$ between two partitions O, S of $\{1, \ldots, d\}$ by $O \leq S$ if S is a sub-partition of O.

Definition 1 (Asymptotic Indepedence-block model)**.** *The random vector* $\boldsymbol{X} = (\boldsymbol{X}^{(1)}, \ldots, \boldsymbol{X}^{(K)})$ *follows a AI-block model if* $\boldsymbol{X}^{(k)} = (X_{i_{k,1}}, \ldots, X_{i_{k,d_k}})$ *are extreme value random vectors for* $k \in \{1, \ldots, K\}$ *and are mutually independent if* $k \neq j$.

Let $\mathcal{L}(\mathbf{X}) = \{O : \mathbf{X} \text{ is AI-block model}\}$ which is nonempty and finite so it does have maximal elements, we note $\mathbf{X} \sim O$ if $\mathbf{X}$ follows an AI-block model. For any partition $O = \{O_k\}_{1 \leq k \leq K}$, we write $j \overset{O}{\sim} k$ if there exist $k \in \{1, \ldots, K\}$ such that $j, k \in G_k$.

Definition 2. *For any two partitions* O, S *of* $\{1, \ldots, d\}$*, we define* $O \cap S$ *as the partition induced by the equivalence relation* $j \overset{O \cap S}{\sim} k$ *iff* $j \overset{O}{\sim} k$ *and* $j \overset{S}{\sim} k$.

Checking that $j \overset{O \cap S}{\sim} k$ is an equivalence relation is straightforward. With this defition, we have the following interesting properties.

Theorem 1. *Let* $\boldsymbol{X}$ *be an extreme value random vector.*

1. *Consider* $O \leq S$*. Then* $\boldsymbol{X} \sim O$ *implies* $\boldsymbol{X} \sim S$.
2. $O \leq O \cap S$ *and* $S \leq O \cap S$
3. *If* $\boldsymbol{X} \sim O$ *or* $\boldsymbol{X} \sim S$*, then* $\boldsymbol{X} \sim O \cap S$.
4. *Thet* $\mathcal{L}(\boldsymbol{X})$ *has a unique maximum* $O^*(\boldsymbol{X})$*, with respect to the partition partial order.*

Proof For the first item, if $\mathbf{X} \sim O$, then the partition O induces mutually independent random vectors. As S is a sub-partition of O, it also generates a partition where vectors are mutually independent.

Now, take $k \in \{1, \ldots, K\}$ and $j, k \in (O \cap S)_k$, we thus have in particular $j \overset{O}{\sim} k$, thus there exists $k' \in \{1, \ldots, K'\}$ such that $a, b \in O'_k$. Thus $(O \cap S)_k \subseteq O_{k'}$. Hence the second statement.

The third result comes down from 1. and 2. We conclude the proof of this propostion by proving the fourth claim. The set $\mathcal{L}(\mathbf{X})$ is non-empty since the trivial partition $O = \{1, \dots, d\}$ belongs to $\mathcal{L}(\mathbf{X})$. It is also a finite set, and we can enumerate it $\mathcal{L}(\mathbf{X}) = \{O_1, \dots, O_m\}$. Define the sequence $O'_1, \dots, O'_M$ recursively according to

- $O'_1 = O_1$,
- $O'_k = O_k \cap O'_{k-1}$ for $k = 2, \dots, M$.

According to Theorem 1, we have that by induction $O'_1, \dots O'_M \in \mathcal{L}(\mathbf{X})$. In addition, we have both $O'_{k-1} \leq O'_k$ and $O_k \leq O'_k$, so by induction $O_1, \dots, O_k \leq O'_k$. Hence the partition $O^*(\mathbf{X}) := O'_M = O_1 \cap \dots \cap O_{M-1}$ is the maximum of $\mathcal{L}(\mathbf{X})$. □

In an AI-block model, we may restrict Equation (2) to the simplex, this equation becomes equivalent to

$$\begin{aligned} A(t_1, \dots, t_d) &= \frac{1}{z_1 + \dots + z_d} \left[\sum_{k=1}^{K} (z_{i_{k,1}} + \dots + z_{i_{k,d_k}}) A^{(k)}(t_{i_{k,1}}, \dots, t_{i_{k,d_k}}) \right] \\ &= \sum_{k=1}^{K} w_k(\mathbf{t}) A^{(k)}(t_{i_{k,1}}, \dots, t_{i_{k,d_k}}), \end{aligned}$$

with $t_j = z_j/(z_1 + \dots + z_d)$ for $j \in 2, \dots, d$ and $t_1 = 1 - t_2 - \dots - t_d$, $w_k(\mathbf{t}) = z_{i_{k,1}} + \dots + z_{i_{k,d_k}}/(z_1 + \dots + z_d)$ for $k \in \{2, \dots, K\}$ and $w_1 = 1 - w_2 - \dots - w_k(\mathbf{t})$ and $t_{i_{k,l}} = z_{i_{k,l}}/(z_{i_{k,1}} + \dots + z_{i_{k,d_k}})$ for $k \in \{1, \dots, K\}$ and $l \in \{1, \dots, d_k\}$. In case of extreme, we thus have in this block independence setting

$$A = A_\Sigma,$$

where $A_\Sigma = \sum_{k=1}^{K} w_k(\mathbf{t}) A^{(k)}(t_{i_{k,1}}, \dots, t_{i_{k,d_k}})$. The function A is still a Pickands dependence function as a convex combination of Pickands dependence function (see p. 123 of [Falk et al., 2010]).

Proposition 1. *In general case, we have for every $\boldsymbol{t} \in S_d$,*

$$(A_\Sigma - A)(\boldsymbol{t}) \geq 0,$$

with equality if and only if $\boldsymbol{X} \sim \mathcal{O}$.

Proof As L is an homogeneous and convex function under a cone, it is also subadditive, *i.e.*

$$L(\mathbf{x} + \mathbf{y}) \leq L(\mathbf{x}) + L(\mathbf{y}),$$

for every $\mathbf{x}, \mathbf{y} \in [0, \infty)^d$. In particular, we obtain

$$L(\sum_{k=1}^{K} \mathbf{z}_k) \leq \sum_{k=1}^{K} L(\mathbf{z}_k),$$

where $\mathbf{x}_k \in [0,\infty)^d$ with $k \in \{1,\dots,K\}$. Consider now $\mathbf{x}_k = (\mathbf{0}, x_{i_{k,1}},\dots,x_{i_{k,d_k}},\mathbf{0})$, we directly obtain

$$L(\mathbf{z}) = L(\sum_{k=1}^{K}\mathbf{z}_k) \leq \sum_{k=1}^{K} L(\mathbf{z}_k) = \sum_{k=1}^{K} L^{(k)}(z_{i_{k,1}},\dots,z_{i_{k,d_k}})$$

Expression this equation in terms of Pickands gives

$$A(\mathbf{t}) \leq \sum_{k=1}^{K} \frac{1}{z_1+\cdots+z_d} L^{(k)}(z_{i_{k,1}},\dots,z_{i_{k,d_k}}) = \sum_{k=1}^{K} \frac{z_{i_{k,1}}+\cdots+z_{i_{k,d_k}}}{z_1+\cdots+z_d} A^{(k)}(t_{i_{k,1}},\dots,t_{i_{k,d_k}}),$$

where $t_i = z_i/(z_1+\cdots+z_d)$. Hence the result.

Another proof is obtained using that extreme value distributions are associated (see Proposition 5.1 of [Marshall and Olkin, 1983] or Section 5.4.1 of [Resnick, 2008]), *i.e.*

$$\mathbb{E}\left[f(\mathbf{X})g(\mathbf{X})\right] \geq \mathbb{E}\left[f(\mathbf{X})\right]\mathbb{E}\left[g(\mathbf{X})\right],$$

for every increasing (or decreasing) functions f, g. By induction, we can prove that, $K \in \mathbb{N}_*$,

$$\mathbb{E}\left[\Pi_{k=1}^{K} f_k(\mathbf{X})\right] \geq \Pi_{k=1}^{K}\mathbb{E}\left[f_k(\mathbf{X})\right] \tag{3}$$

Take $f_k(\mathbf{x}) = \mathbb{1}_{\{]-\infty,\mathbf{x}_k\}}$ for each $k \in \{1,\dots K\}$, thus Equation (3) gives

$$C(G_1(x_1),\dots,G_d(x_d)) \geq \Pi_{k=1}^{K} C_k\left(G_{i_{k,1}}(x_{i_{k,1}}),\dots,G_{i_{k,d_k}}(x_{i_{k,d_k}})\right),$$

which can be restated in terms of stable tail dependence function

$$L(\mathbf{z}) \leq \sum_{k=1}^{K} L^{(k)}(z_{i_{k,1}},\dots,z_{i_{k,d_k}}),$$

where $\mathbf{z}_k \in [0,\infty)^{d_k}$ and $\mathbf{z} = (\mathbf{z}_1,\dots,\mathbf{z}_K)$. We obtain the statement expressing this inequality with Pickands dependence function.

For the rest of the proof, notice that (3) holds as an inequality if and only if $\mathbf{X}^{(1)} \perp\!\!\!\perp \dots \perp\!\!\!\perp \mathbf{X}^{(K)}$. $\square$

3 The similarity metric and its induced cluster separation

Classical clustering methods, such as K-means and hierarchical clustering, typically build clusters for which the within-group correlation is larger than that between groups. If $X \sim \mathcal{O}$ has a block structure with associated Pickands A which can be expressed as a convex combination of the K pickands $A^{(k)}$. One expects that consistent clustering couldbe possible when

$$A_\Sigma - A \tag{4}$$

is small enough. To motivate this metric, notice that in AI-block models, if a given partition belongs indeed to $\mathcal{O}$, then (4) is equal to 0. This statement is summarized below.

Proposition 2. *Let $\boldsymbol{X} \sim \mathcal{L}(\boldsymbol{X})$. Let us consider $S_l = \{j_{l,1}, \dots, j_{l,D_l}\}$ an arbitrary partition of $\{1, \dots, d\} = \bigsqcup_{l=1}^{L} S_l$ with $D_1 + \cdots + D_L = d$. We thus have :*

$$0 = (A_\Sigma - A)(\boldsymbol{t}) \leq \sum_{l=1}^{L} \sum_{k=1}^{K} w_{kl} A^{(k)}(\boldsymbol{0}, t_{kl}, \boldsymbol{0}) - A(\boldsymbol{t}), \quad \forall \boldsymbol{t} \in S_d, \tag{5}$$

where by denoting $\boldsymbol{z} \in [0, \infty)^d$,

$$\begin{aligned}
w_{kl} &= \sum_{j \in S_l \cap O_k} \frac{z_j}{z_1 + \cdots + z_d}, \quad \forall k \in \{1, \dots, K\},\, l \in \{1, \dots, L\}, \\
t_{kl} &= \left(\frac{z_i}{\sum_{j \in S_l \cap O_k} z_j} \right)_{i \in S_l \cap O_k}, \quad \forall k \in \{1, \dots, K\},\, l \in \{1, \dots, L\}, \\
t_j &= \frac{z_j}{z_1 + \cdots + z_d}, \quad j \in \{1, \dots, d\}.
\end{aligned}$$

Furthermore, the right hand side of the inequality is equal to zero if and only if $S \leq O$.

Proof Note that $\mathbf{Y}^{(l)} = (\mathbf{X}_{j_{l,1}}, \dots, \mathbf{X}_{j_{l,D_l}})$ and $\mathbf{Y}^{(m)} = (\mathbf{X}_{j_{m,1}}, \dots, \mathbf{X}_{j_{m,D_m}})$ are not, in general independent where $l, m \in \{1, \dots L\}$ because the partition $\bigsqcup_{l=1}^{L} S_l$ is arbitrarily given. Consider for some $l \in \{1, \dots, L\}$,

$$f_l = \mathbb{1}_{\{]-\infty, \mathbf{x}_l\}}, \quad \mathbf{x}_l = (x_{j_{l,1}}, \dots, x_{j_{l,D_l}}).$$

We thus have in one hand

$$\mathbb{E}\left[\Pi_{l=1}^{L} f_l(\mathbf{X})\right] = \mathbb{P}\left\{X_1 \leq x_1, \dots, X_d \leq x_d\right\} = \Pi_{k=1}^{K} C_k \left(G_{i_{k,1}}(x_{i_{k,1}}), \dots, G_{i_{k,d_k}}(x_{i_{k,d_k}}) \right),$$

on the other hand

$$\Pi_{l=1}^{L} \mathbb{E}\left[f_l(\mathbf{X})\right] = \Pi_{l=1}^{L} \mathbb{P}\left\{X_{j_{l,1}} \leq x_{j_{l,1}}, \dots, X_{j_{l,D_l}} \leq x_{j_{l,D_l}}\right\}.$$

It is worth noticing that the above set can be rewriten as

$$\left\{X_{j_{l,1}} \leq x_{j_{l,1}}, \dots, X_{j_{l,D_l}} \leq x_{j_{l,D_l}}\right\} = \bigcap_{k=1}^{K} \bigcap_{i \in S_l \cap O_k} \left\{X_i \leq x_i\right\}.$$

Notice that $\bigcap_{i \in \emptyset} \{X_i \leq x_i\} = \Omega$. As a subvector of $\mathbf{X}^{(k)}$, $\mathbf{X}_{kl} = (X_i)_{i \in S_l \cap O_k}$ is independent of $\mathbf{X}^{(j)}$ for $j \neq k$. Using this block-independence, we thus obtain :

$$\mathbb{P}\left\{X_{j_{l,1}} \leq x_{j_{l,1}}, \dots, X_{j_{l,D_l}} \leq x_{j_{l,D_l}}\right\} = \Pi_{k=1}^{K} \mathbb{P}\left\{\bigcap_{i \in S_l \cap O_k} \left\{X_i \leq x_i\right\}\right\}.$$

Now using positive association, we obtain that

$$\Pi_{k=1}^{K} C_k \left(G_{i_{k,1}}(x_{i_{k,1}}), \ldots, G_{i_{k,d_k}}(x_{i_{k,d_k}}) \right) \geq \Pi_{l=1}^{L} \Pi_{k=1}^{K} C_k \left(\mathbf{1}, \mathbf{x}_{kl}, \mathbf{1} \right),$$

where $\mathbf{x}_{kl} = (x_i)_{i \in S_l \cap O_k}$. Reexpressing the whole in terms of stable tail dependence function leads to :

$$\sum_{k=1}^{K} L^{(k)}(z_{i_{k,1}}, \ldots, z_{k,d_k}) \leq \sum_{l=1}^{L} \sum_{k=1}^{K} L^{(k)}(\mathbf{0}, \mathbf{z}_{kl}, \mathbf{0}),$$

with $z_i = -\ln(G_i(x_i))$ for every $i \in \{1, \ldots, d\}$. Dividing by $z_1 + \cdots + z_d$ gives

$$A_\Sigma(\mathbf{t}) = \sum_{k=1}^{K} w_k(\mathbf{t}) A^{(k)}(t_{i_{k,1}}, \ldots, t_{i_{k,d_k}}) \leq \sum_{l=1}^{L} \sum_{k=1}^{K} \frac{1}{z_1 + \cdots + z_d} L^{(k)}(\mathbf{0}, \mathbf{z}_{kl}, \mathbf{0}).$$

The right hand side of the equation can be written as $\forall k \in \{1, \ldots, K\}$, $\forall l \in \{1, \ldots, L\}$

$$\begin{aligned} \frac{1}{z_1 + \cdots + z_d} L^{(k)}(\mathbf{0}, z_{kl}, \mathbf{0}) &= \frac{\sum_{j \in S_l \cap O_k} z_j}{z_1 + \cdots + z_d} L^{(k)} \left(\mathbf{0}, \frac{\mathbf{z}_{lk}}{\sum_{j \in S_l \cap O_k} z_j}, \mathbf{0} \right) \\ &\triangleq w_{kl} A^{(k)}(\mathbf{0}, \mathbf{t}_{kl}, \mathbf{0}). \end{aligned}$$

We thus obtain the statement.

Now, if $\forall k \in \{1, \ldots, K\}$, $\exists l \in \{1, \ldots, L\}$ such that $O_k \subseteq S_l$, then the random vectors $\mathbf{Y}^{(l)} = (\mathbf{X}_{j_{l,1}}, \ldots, \mathbf{X}_{j_{l,D_l}})$ and $\mathbf{Y}^{(m)} = (\mathbf{X}_{j_{m,1}}, \ldots, \mathbf{X}_{j_{m,D_m}})$ are now independent. If we suppose $L \geq K$, we thus have $S_1 = O_1, \ldots, S_K = O_K$ and $S_l = \emptyset$ for $l > K$. The equality comes down from Lemma 1. Now, if $L < K$, we have with the same notations in the proof

$$\mathbb{E}\left[\Pi_{l=1}^{L} f_l(\mathbf{X})\right] = \Pi_{l=1}^{L} \mathbb{P}\left\{ X_{j_{l,1}} \leq x_{j_{l,1}}, \ldots, X_{j_{l,D_l}} \leq x_{j_{l,D_l}} \right\} = \Pi_{k=1}^{K} \mathbb{P}\left\{ \mathbf{X}^{(k)} \leq \mathbf{x}_k \right\}.$$

Expressing this two terms as before, we obtain that $A = A_\Sigma$. For the converse, suppose that the right hand side of the inequality in (5) is equal to zero. Applying Lemma 1 gives that the arbitrary partition has the same value as A_Σ. That is saying that the random vectors inside $S_1, \ldots S_L$ are mutually independent. If $L \geq K$, thus, apart for the K first clusters say for which $S_k = O_k$, the others are empty set. Now, if $L < K$, we group one or more O_k in a given cluster S_l without one overlaps to an other cluster S_j say (if it does, the value could not be equal as zero by Lemma 1). Hence the statement. □

Proposition 2 states that the (unique up to a permutation) partitioning $\mathbf{O} = \{O_1, \ldots, O_K\}$ of the set $\{1, \ldots, d\}$, also called the ground-truth clustering, equals 0 and is lower or equal than any given arbitrary partition of the set S. Also, if the criteria is equal to 0, we indeed have a independent partition but it doesn't say more if it is possible or no to obtain a finer partition. Also, this criteria if useful to recover clusters, as given by the following proposition.

Proposition 3 (Recover of clusters)**.** *Under the hypothesis of Proposition 2, we thus have , for every* $k \in \{1, \dots, K\}$

$$w_k(\boldsymbol{t})A^{(k)}(t_{i_{k,1}}, \dots, t_{i_{k,d_k}}) + \sum_{l=1}^{L-1} \sum_{j=1, j\neq k}^{K} w_{jl} A_j(\boldsymbol{0}, t_{jl}, \boldsymbol{0}) - A(\boldsymbol{t}) \leq \sum_{l=1}^{L} \sum_{k=1}^{K} w_{kl} A^{(k)}(\boldsymbol{0}, t_{kl}, \boldsymbol{0}) - A(\boldsymbol{t}),$$

for every $t \in S_d$ *and where we arbitrarily set the Lth cluster as* $O_k = \{i_{k,1}, \dots, i_{k,d_k}\}$

Proof This directly comes down from the subadditivy of the stable tail dependence function. Without loss of generality, set $k = 1$, and consider $O_1 = \bigsqcup_{l=1}^{L} S_l$ a partition of O_1 with $\mathbf{z}_l = (z_i)_{i \in S_l}$ for every $n \in \{1, \dots, N\}$. We thus have $(z_{i_{1,1}}, \dots, z_{i_{k,d_1}}) = \sum_{l=1}^{L} (\mathbf{0}, \mathbf{z}_l, \mathbf{0})$. Using subadditivity gives :

$$L_1(z_{i_{1,1}}, \dots, z_{i_{1,d_1}}) \leq \sum_{l=1}^{L} L_1(\mathbf{0}, \mathbf{z}_l, \mathbf{0}) = \sum_{l=1}^{L} L_1(\mathbf{0}, \mathbf{z}_{kl}, \mathbf{0}).$$

Adding the other clusters, expressing the whole as Pickands dependence function and substract by A gives the result. □

Proposition 3 tells us that, for a given group of variable $k \in \{1, \dots, K\}$, the lower bound is attained when each element of this group are clustered together.

Furthermore, in order to build up cluster, one may be interested by the following metric. For notational convenience, we note by $O'_k = \{i_{k,1}, \dots, i_{k,L}\}$ with $L \leq d_1$.

$$Pick(j, O'_k) = R^{(k)}(t_{i_{k,1}}, \dots, t_{i_{k,L}}) + t_j - R^{(k)}(t_{i_{k,1}}, \dots, t_{i_{k,L}}, t_j), \tag{6}$$

where

$$R^{(k)}(t_{i_{k,1}}, \dots, t_{i_{k,L}}) = w_k^L(t_{i_{k,1}}, \dots, t_{i_{k,L}}) A^{(k)} \left(\frac{t_{i_{k,1}}}{w_k^L(t_{i_{k,1}})}, \dots, \frac{t_{i_{k,L}}}{w_k^L(t_{i_{k,1}})}, \mathbf{0} \right),$$

and $w_k^L(t_{i_{k,1}}, \dots, t_{i_{k,L}}) = \sum_{j \in O'_k} t_j$. To motivate this metric, in AI-block model, a variable with indice j is in the same group of variables with indices $\{i_{k,1}, \dots, i_{k,L}\}$, then (6) is strictly positive. Conversely, if $i_{k,L+1}$ is not in the same group of $\{i_{k,1}, \dots, i_{k,L}\}$, then (6) is equal to 0. This motivates our metric which is tailored to AI-block model. This discussion is detailed by the following proposition.

Proposition 4 (We live in a society)**.** *Let* $\boldsymbol{t} \in S_{d_k}$, $\boldsymbol{X}^{(k)}$ *an extremal random vector in* $\mathbb{R}^{d_k}$ *and* $A^{(k)}$ *its associated Pickands dependence function, we have*

$$\begin{aligned} A^{(k)}(t_{i_{k,1}}, \dots, t_{i_{k,d_k}}) &\leq \sum_{j=1}^{d_k-1} t_{i_{k,j}} A^{(k)} \left(\frac{t_{i_{k,1}}}{\sum_{j=1}^{d_k-1} t_{i_{k,j}}}, \frac{t_{i_{k,2}}}{\sum_{j=1}^{d_k-1} t_{i_{k,j}}}, \dots, \frac{t_{i_{k,d_k-1}}}{\sum_{j=1}^{d_k-1} t_{i_{k,j}}}, 0 \right) + t_{i_{k,d_k}} \\ &\leq \dots \leq \sum_{j=1}^{2} t_{i_{k,j}} A^{(k)} \left(\frac{t_{i_{k,1}}}{\sum_{j=1}^{2} t_{i_{k,j}}}, \frac{t_{i_{k,2}}}{\sum_{j=1}^{2} t_{i_{k,j}}} 0, \dots, 0 \right) + t_{i_{k,3}} + \dots + t_{i_{k,d_k}} \leq 1, \end{aligned}$$

where inequalities holds if and only if $\boldsymbol{X}^{(k)}$ is totally independent.

Proof This is straightforward from the subadditivity of the stable tail dependence function, indeed, for $\mathbf{z} \in \mathbb{R}_+^d$

$$L(z_1, \ldots, z_d) = L((z_1, \ldots, z_{d-1}, 0) + (0, \ldots, 0, z_d)) \leq L(z_1, \ldots, z_{d-1}, 0) + L(0, \ldots, 0, z_d). \quad (7)$$

And notice that $L(0, \ldots, 0, z_d) = z_d$ by definition of the stable tail dependence function and the condition of the spectral measure. Proceed recursively on $L(z_1, \ldots, z_{d-1}, 0)$ gives the other inequalites. Expressing the whole as Pickands states the result.

Now, if $\mathbf{X}^{(k)}$ is totally independent, we have that $L(z_{i_{k,1}}, \ldots, z_{i_{k,d_k}}) = z_{i_{k,1}} + \cdots + z_{i_{k,d_k}}$ and the equalities follows immediately. For the converse, Equation (7) states as an equality, proceed recursively and we obtain that :

$$L(z_{i_{k,1}}, \ldots, z_{i_{k,d_k}}) = z_{i_{k,1}} + \cdots + z_{i_{k,d_k}}.$$

hence the statement. □

An other interpretation of the second conclusion of Proposition 4 is that the only stable tail dependence function which are linear is $L(z_1, \ldots, z_d) = z_1 + \cdots + z_d$.

4 An algorithm for variable clustering with known distribution

In this section, $\mathbf{X} \in \mathbb{R}^d$ is a random vector whose the pickands dependence function A is known. The access to this distribution is via an oracle given in Section that is

$$crit(S_1, S_2) = A - A_{(S_1,S_2)}, \quad (8)$$

with $A_{(S_1,S_2)}(\mathbf{t}) = w_1(\mathbf{t})A_1(t_{i_{1,1}}, \ldots, t_{i_{1,d_1}}) + w_2(\mathbf{t})A_2(t_{i_{2,1}}, \ldots, t_{i_{2,d_2}})$ and $S_2 = \{i_{1,1}, \ldots, i_{1,d_1}\}$, $S_1 = \{i_{2,1}, \ldots, i_{2,d_2}\}$ are two index sets where $d_1 + d2 = d$. According to Proposition 2, the oracle will provide answers for given disjoints sets $S_1, S_2 \in S$ of random variables if it is an AI-block model. Furthermore, the statistician has access to the other oracle function whose definition is recall below

$$Pick(j, O_1') = R^{(1)}(t_{i_{1,1}}, \ldots, t_{i_{1,L}}) + t_j - R^{(1)}(t_{i_{1,1}}, \ldots, t_{i_{1,L}}, t_j)$$

which is strictly positive if the variable with indice $j \in O \setminus O_1'$ indeed belong to O_1' and equals to 0 if independent of O_1'.

The proposed algorithm (see the pseudocode listing below) works as follows. It attempts to split the input set recursively into two independent clusters, until it is no longer possible. To split a set in two, it starts with a candidate cluster $C = \{1\}$ and measure its discrepancy between the set $R = \bigcup_{j=2}^d S_j$. If $crit(C, R)$ is already 0, then we have split the set into two independent clusters

and can stop. Otherwise, the algorithm then takes element from R one by one without replacement and shift its position while looking whether the criteria $Pick$ is positive. Once such an element is found, it is moved into the other cluster.

Theorem 2. *The algorithm outputs the correct clustering at $\mathcal{O}((K-1)d^2)$ oracle calls.*

Proof Adapted from Theorem 1 of [Ryabko, 2017]

We shall first show that the procedure for splitting a set into two indeed splits set into two independent sets, if and only if such two sets exists. First, if $crit(C, R) = 0$, then $C \perp\!\!\!\perp R$ and the function terminates. In the opposite case, when $crit(C, R) > 0$, by Proposition 4, there exist an element $i \in S$ such that a shifted position would lead to a decrease of the criteria $Pick(j, C)$. This process will continue as long as one element is not attached to its original cluster by Proposition 3. Since there are only finitely many elements in S, the while loop eventually terminates at d iteration. Finally, notice that if $(C_1, C_2) \perp\!\!\!\perp C_3$ and $C_1 \perp\!\!\!\perp C_2$ then also $C_1 \perp\!\!\!\perp C_2 \perp\!\!\!\perp C_3$, which means that by repeating the Split function recursively, we find the correct clustering. □

Algorithm 1 Recursive algorithm to cluster with K unknown

```
Data: The set S
Result: Cluster with K unknown, given an oracle
procedure ALG(S)
    (S_1, S_2) = Split(S)
    if S_2 ≠ ∅ then
        return ALG(S_1), ALG(S_2)
    else
        return S_1
procedure SPLIT(S)
    Initialize : C := {1}, R := {2, ..., d}.
    while crit(C, R) > 0 do
        for j ∈ R do
            if Pick(j, C) > 0 then
                move j from R to C
            else
                j stays in R
    return C, R
```

5 Estimation of Cluster

2.1- Statistical inference In this section, we propose a estimation approch that utilizes non-parametric estimation of the Pickands dependence function and we use it for inference to recover the cluster.

Let $\mathbf{X}_1, \dots \mathbf{X}_n$ be *i.i.d.* copies of $\mathbf{X}$. This estimator is based on the madogram concept, a notion borrowed from geostatistics in order to capture the spatial strucure. Our estimator is defined as

$$\hat{A}_n(\mathbf{t}) = \frac{\hat{\nu}_n(\mathbf{t}) + c(\mathbf{t})}{1 - \hat{\nu}_n(\mathbf{t}) - c(\mathbf{t})}, \tag{9}$$

where

$$\hat{\nu}_n(\mathbf{t}) = \frac{1}{n}\sum_{j=1}^{n}\left[\bigvee_{j=1}^{d}\{G_{n,j}(X_{i,j})\}^{1/t_j} - \frac{1}{d}\sum_{j=1}^{d}\{G_{n,j}(X_{i,j})\}^{1/t_j}\right],$$
$$c(\mathbf{t}) = \frac{1}{d}\sum_{j=1}^{d}\frac{t_j}{1+t_j}.$$

by convention, here $u^{1/0} = 0$ for $u \in (0,1)$. We state a concentration inequality below for this statistic.

Lemma 2. *For any $\boldsymbol{t} \in S_d$ and $\delta \in]0,1[$, with probability larger than $1-\delta \in]0,1[$,*

$$\left|\hat{A}_n(\boldsymbol{t}) - A(\boldsymbol{t})\right| < 2\sqrt{\frac{2}{n}\ln\left(\frac{4d}{\delta}\right)}.$$

Proof Remark that the function $f(x) = x/(1+x)$ for $x \in \mathbb{R}_+$ is 1-Lipschitz, thus for every $\mathbf{t} \in S_d$:

$$|\hat{A}_n(\mathbf{t}) - A(\mathbf{t})| = |f(\hat{\nu}_n(\mathbf{t})) - f(\nu(\mathbf{t}))| \leq |\hat{\nu}_n(\mathbf{t}) - \nu(\mathbf{t})|.$$

We thus have, for $\epsilon > 0$,

$$\mathbb{P}\left\{|\hat{A}_n(\mathbf{t}) - A(\mathbf{t})| > \epsilon\right\} \leq \mathbb{P}\left\{|\hat{\nu}_n(\mathbf{t}) - \nu(\mathbf{t})| > \epsilon\right\}. \tag{10}$$

Now, observe that

$$|\hat{\nu}_n(\mathbf{t}) - \nu(\mathbf{t})| \leq |\hat{\nu}_n(\mathbf{t}) - \nu_n(\mathbf{t})| + |\nu_n(\mathbf{t}) - \nu(\mathbf{t})|,$$

where

$$\nu_n(\mathbf{t}) := \sum_{i=1}^{n} Y_i = \sum_{i=1}^{n}\frac{1}{n}\left[\bigvee_{j=1}^{d}\{F_j(X_{i,j})\}^{1/t_j} - \frac{1}{d}\sum_{j=1}^{d}\{F_j(X_{i,j})\}^{1/t_j}\right].$$

As the following inequalities holds for every $i \in \{1, \dots, n\}$

$$Y_i \leq \frac{(d-1)}{dn}.$$

Hoeffding's inequality applies and we obtain that

$$\mathbb{P}\left\{|\nu_n(\mathbf{t}) - \nu(\mathbf{t})| > \epsilon/2\right\} \leq 2\exp\left(-\frac{nd^2\epsilon^2}{4(d-1)^2}\right).$$

Furthermore, we have

$$|\hat{\nu}_n(\mathbf{t}) - \nu_n(\mathbf{t})| \leq 2 \sup_{j\in\{1,\dots,d\}} \sup_{i\in\{1,\dots,n\}} \left| \left\{\hat{F}_{n,j}(X_{i,j})\right\}^{1/t_j} - \{F_j(X_{i,j})\}^{1/t_j} \right|$$

Applying DKW inequality, we obtain

$$\mathbb{P}\left\{ \sup_{j\in\{1,\dots,d\}} \sup_{i\in\{1,\dots,n\}} \left| \left\{\hat{F}_{n,j}(X_{i,j})\right\}^{1/t_j} - \{F_j(X_{i,j})\}^{1/t_j} \right| > \frac{\epsilon}{4} \right\} \leq 2d \exp\left(\frac{-n\epsilon^2}{8}\right).$$

We thus have for $d \geq 2$

$$\mathbb{P}\left\{|\hat{\nu}_n(\mathbf{t}) - \nu(\mathbf{t})| > \epsilon\right\} \leq 4d \exp\left(-\frac{n\epsilon^2}{8}\right).$$

Let $\delta \in]0,1[$, solve the following equation with respect to ϵ

$$\frac{\delta}{4} = d \exp\left(-\frac{n\epsilon^2}{8}\right).$$

and use Equation (10) gives the result. □

In order to build a new test of stochastic vectorial asymptotic independence, fix $k \in \{1,\dots,K\}$ and let $\hat{A}_{nk}(t_{i_{k,1}},\dots,t_{i_{k,d_k}}) = \hat{A}_n(\mathbf{0}, t_{i_{k,1}},\dots,t_{i_{k,d_k}},\mathbf{0})$ denotes the empirical Pickands dependence function associated to the k-th subvector of X where $\mathbf{0} = (0,\dots,0)$. Then consider the empirical process

$$\mathcal{E}_{nK}(\mathbf{t}) = \sqrt{n}\left(\hat{A}_n(\mathbf{t}) - \hat{A}_{n\Sigma}(\mathbf{t})\right), \tag{11}$$

where $\hat{A}_{n\Sigma} = \sum_{k=1}^{K} w_k(\mathbf{t}) \hat{A}_{nk}(t_{i_{k,1}},\dots,t_{i_{k,d_k}})$.

Theorem 3. *Under $\mathcal{H}_0$, the empirical process $\mathcal{E}_{nK}$ converges weakly in $\ell^\infty(S_d)$ to a tight Gaussian process having representation*

$$\begin{aligned}\mathcal{E}_K(\boldsymbol{t}) = &-(1+A(\boldsymbol{t}))^2 \int_{[0,1]} N_C(u^{t_1},\dots,u^{t_d})du \\ &+ \sum_{k=1}^{K} w_k(\boldsymbol{t}) \left(1 + A^{(k)}(t_{i_{k,1}},\dots,t_{i_{k,d_k}})\right)^2 \int_{[0,1]} N_C(\boldsymbol{1}, u^{t_{i,1}},\dots,u^{t_{i,d_k}},\boldsymbol{1})du,\end{aligned}$$

where N_C is a continuous tight Gaussian process with representation

$$N_C(u_1,\dots,u_d) = B_C(u_1,\dots,u_d) - \sum_{j=1}^{d} \dot{C}_j(u_1,\dots,u_d) B_C(\boldsymbol{1}, u_i, \boldsymbol{1}),$$

and B_C is a continuous tight Gaussian process with covariance function

$$cov(B_C(\boldsymbol{u}), B_C(\boldsymbol{v})) = C(\boldsymbol{u}\wedge\boldsymbol{v}) - C(\boldsymbol{u})C(\boldsymbol{v}) \overset{\boldsymbol{X}\sim\mathcal{L}(\boldsymbol{X})}{=} C_\Pi(\boldsymbol{u}\wedge\boldsymbol{v}) - C_\Pi(\boldsymbol{u})C_\Pi(\boldsymbol{v})$$

$$\begin{tikzcd}
\sqrt{n}\left(\hat{A}_n - A\right) \arrow[r] \arrow[d, "\phi"'] & \mathcal{E}_{nK} \\
\left(\sqrt{n}\left(\hat{A}_n - A\right); w_1\sqrt{n}\left(\hat{A}_{n1} - A_1\right), \ldots, w_k(\mathbf{t})\sqrt{n}\left(\hat{A}_{nk} - A^{(k)}\right)\right) \arrow[ur, "\psi"'] &
\end{tikzcd}$$

Figure 1: Diagram of composition of function.

Proof The proof is straightforward, notice that (see Figure 1)

$$\mathcal{E}_{nK} = \psi \circ \phi\left(\sqrt{n}(\hat{A}_n - A)\right),$$

where ϕ is detailed as

$$\begin{array}{rcccl} \phi & : & \ell^\infty(S_d) & \to & \ell^\infty(S_d) \otimes (\ell^\infty(S_d), \ldots, \ell^\infty(S_d)) \\ & & x & \mapsto & (x, \phi_1(x), \ldots, \phi_K(x)), \end{array}$$

with for every $k \in \{1, \ldots, K\}$

$$\begin{array}{rcccl} \phi_k & : & \ell^\infty(S_d) & \to & \ell^\infty(S_d) \\ & & x & \mapsto & x(\mathbf{0}, t_{i_{k,1}}, \ldots, t_{i_{k,d_k}}, \mathbf{0}). \end{array}$$

Thus ϕ_k is a linear and bounded function hence continuous, it follows that ϕ is continuous since each coordinate functions are continuous. Using that (see [Marcon et al., 2017])

$$\sqrt{n}(\hat{A}_n(\mathbf{t}) - A(\mathbf{t})) \rightsquigarrow -(1 + A(\mathbf{t}))^2 \int_{[0,1]} N_C(u^{t_1}, \ldots, u^{t_d}) du,$$

and applying the continuous mapping theorem for the weak convergence in $\ell^\infty(S_d)$ (Theorem 1.3.6 of [van der Vaart et al., 1996]) leads the result. □

Remark 2. *If we want to test the independence between the marginals* within *the vector of* $\boldsymbol{X}$*, we set* $A_1 = \cdots = A^{(k)} = 1$ *and thus* $A_\Sigma = \sum_{k=1}^K w_k(\boldsymbol{t}) = 1$*. The null and the alternative hypothesis can thus be written as*

$$\mathcal{H}_0 : A = 1, \qquad \mathcal{H}_1 : A \neq 1.$$

We return to the set of hypothesis as stated by [Guillou et al., 2018] and the result obtained in Theorem 3 applies in this framework, indeed, remark that for $d_1 = \cdots = d_K = 1$ *we have that :*

$$\mathbb{P}\left\{u \in [0,1],\ N_C(\boldsymbol{1}, u, \boldsymbol{1}) = 0\right\} = 1.$$

This result follows from the covariance function of N_C *and the continuity of its trajectories. We*

thus obtain the following convergence :

$$\sqrt{n}(\hat{A}_n(\boldsymbol{t}) - 1) \rightsquigarrow -4 \int_{[0,1]} N_C(u^{t_1}, \ldots, u^{t_d}) du, \tag{12}$$

that is the asymptotic result used in [Guillou et al., 2018] which is a specific case for our framework if we are interested in testing the asymptotic independence between random vectors.

2.2- Consistent estimation of clusters In this paragraph, we present an algorithm that recover clusters in AI-block models. We recal that we place j in the group $O'_k = \{i_{k,1}, \ldots, i_{k,L}\}$ if $Pick(j, G'_k) > 0$, the latter happens if and only if j indeed belong to G'_k. As we do not have knowledge of A, we need to estimate it from n observed independent copy $\mathbf{X}_1, \ldots, \mathbf{X}_n$ of $\mathbf{X}$. To a given estimator $\hat{A}_n$ of A we associate the estimator

$$\hat{R}_n(t_{i_{1,1}}, \ldots, t_{i_{1,L}}) = w_k^L(\mathbf{t}) \hat{A}_n^{(1)} \left(t_{i_{1,1}}, \ldots t_{i_{1,L}}, \mathbf{0}\right).$$

Using this estimator, we define

$$\widehat{Pick}(j, O'_1) = \hat{R}_n(t_{i_{1,1}}, \ldots, t_{i_{1,L}}) + t_j - \hat{R}_n(t_{i_{1,1}}, \ldots, t_{i_{1,L}}, t_j). \tag{13}$$

of the $Pick$ metric. We then estimate the partition $\hat{O}$ according to the following algorithm.

Algorithm 2 Split procedure with A unknown

```
procedure SPLIT(S, τ, R̂_n)
    Initialize : C := {1}, R := {2, ..., d}.
    while crit(C, R) strictly decreases do
        for j ∈ R do
            if Pick(j, C) > τ_{j,C} then
                move i from R to C
            else
                i stays in R
    return C, R
```

In the following, we provide conditions ensuring that $\hat{O} = O$. Without lose of generality, we suppose that $i_{1,1} = \{1\} \in O_1$.

Theorem 4. *Set $\boldsymbol{t} \in S_d$. Consider the AI-block model with $d \geq 2$ and $K \leq d$. Let $0 < \delta < 1$, $U \subseteq \{1, \ldots, d\}$ and define $\tau_{a,U} = C(2w_1^U(\boldsymbol{t}) + t_a)r$ where $a \in S \setminus U$ for some $C \geq 1$, then with probability greater than $1 - \delta$, we have*

$$\hat{O}_1 \subseteq O_1$$

where

$$O_1 = \bigcup_{k=1}^{d} O_1^k, \quad O_1^k = \inf\left\{j \in O \setminus O_1^{k-1}, Pick(j, O_1^{k-1}) > 0\right\}, \quad O_1^1 = \{1\}.$$

and

$$\hat{O}_1 = \bigcup_{k=1}^{d} \hat{O}_1^k, \quad \hat{O}_1^k = \inf\left\{ j \in O \setminus \hat{O}_1^{k-1}, \widehat{Pick}(j, \hat{O}_1^{k-1}) > \tau_{j,\hat{O}_1^{k-1}} \right\}, \quad \hat{O}_1^1 = \{1\}.$$

Define $\tau_{j,O_1^{k-1}} = 2(w_1^{O_1^{k-1}}(\boldsymbol{t}) + t_j)r$ *and assume that*

$$\min_{j \in O_1 \setminus O_1^{k-1}} Pick(j, O_1^{k-1}) > 2\tau,$$

then

$$\mathbb{P}\left\{ \hat{O}_1 = O_1 \right\} \geq 1 - \delta$$

Proof Set $r_{|U|} = 2\sqrt{\frac{2}{n} \ln\left(\frac{4|U|}{\delta}\right)}$. Consider the random event

$$\mathcal{A}^k = \left\{ \mathbf{t} \in S_{d_k}, \left| \hat{A}_n^U(\mathbf{t}) - A^U(\mathbf{t}) \right| < r_k \right\}, \quad k \in \{1, \ldots, d\}, \quad U \subseteq \{1, \ldots, d\}, \quad |U| = k,$$

with $\hat{A}_n^U$ is the estimator of the pickands dependence function A^U computed for the variables X_j with indices j in U. By Lemma 2,

$$\mathbb{P}\left((\cup_{k=1}^d A^k)^c \right) = \mathbb{P}((\mathcal{A}^d)^c) \leq \delta,$$

and thus $\mathbb{P}(\cap_{k=1}^d A^k) \geq 1 - \delta$. We now show that, on the event $\mathcal{A}^d$,

$$\hat{O}_1 \subseteq O_1.$$

For $k = 1$, we have $\hat{O}_1^1 = \{1\} \subseteq O_1^1 = \{1\}$. For $k = 2$, suppose, without loss of generality that $2 \in \hat{O}_1$. Then

$$\widehat{Pick}(j, \hat{O}_1^1) = (t_1 + t_2)\left(1 - \hat{A}_n^{(1)}\left(\frac{t_1}{t_1 + t_2}, \frac{t_2}{t_1 + t_2}, \mathbf{0}\right)\right) > \tau_{2,\hat{O}_1^1} = C(t_1 + t_2)r_2.$$

We have

$$\widehat{Pick}(2, \hat{O}_1^1) = \left[\widehat{Pick}(2, \hat{O}_1^1) - Pick(2, \hat{O}_1^1)\right] + Pick(2, \hat{O}_1^1).$$

Observe now that the term under brackets is equal too

$$(t_1 + t_2)(A^{(1)} - \hat{A}_n^{(1)})\left(\frac{t_1}{t_1 + t_2}, \frac{t_2}{t_1 + t_2}, \mathbf{0}\right).$$

Using that, on $\mathcal{A}^2$,

$$(A^{(1)} - \hat{A}_n^{(1)})\left(\frac{t_1}{t_1 + t_2}, \frac{t_2}{t_1 + t_2}, \mathbf{0}\right) < r_2,$$

we thus obtain

$$Pick(2, \hat{O}_1^2) > \tau_{2,\hat{O}_1^1} - (t_1 + t_2)r_2 \geq 0.$$

This ensures that $2 \in O_1^2$. Suppose now that $k \leq d$, without loss of generality suppose that $k \in \hat{O}_1^k$ and proceeding as below, we have

$$\widehat{Pick}(k, \hat{O}_1^{k-1}) = \left[\widehat{Pick}(k, \hat{O}_1^{k-1}) - Pick(k, \hat{O}_1^{k-1})\right] + Pick(k, \hat{O}_1^{k-1}).$$

Notice that the term in bracket is strictly lower than $r_k(2w_k^{\hat{O}_1^{k-1}} + t_k)$, and as $k \in \hat{O}_1^k$, we obtain that

$$Pick(k, \hat{O}_1^{k-1}) > \tau_{k,\hat{O}_1^{k-1}} - r_k\left(2w_k^{\hat{O}_1^{k-1}} + t_k\right) > 0.$$

We thus obtain that $k \in O_1^k$ for $k \leq d$. Using finite union over $k \in \{1, \ldots, d\}$, we obtain

$$\hat{O}_1 \subseteq O_1$$

We now prove the second part of the theorem. We need to prove that $\hat{O}_1 = O_1$ on $\cap_{k=1}^d \mathcal{A}^k$ when $\tau_{j,O_k^1} = 2r_k$. We already have with the first part of the theorem $\hat{O}_1 \subseteq O_1$. Thus it remains to show that $O_1 \subseteq \hat{O}_1$. For $k = 1$, it is trivial, let take $k = 2$ and the general case with $k \leq d$ follows similarly from it. Indeed, assume that $2 \in O_1$. Then, by the assumptions of the theorem

$$Pick(2, O_1^1) > 2\tau_{j,O_1^1}.$$

Using this property, we have

$$(t_1 + t_2)(1 - \hat{A}_n^{(1)})\left(\frac{t_1}{t_1 + t_2}, \frac{t_2}{t_1 + t_2}, \mathbf{0}\right) > (t_1 + t_2)(1 - A^{(1)})\left(\frac{t_1}{t_1 + t_2}, \frac{t_2}{t_1 + t_2}, \mathbf{0}\right) - \tau_{j,O_1^1} > \tau_{j,O_1^1}.$$

Thus, we do have $2 \in \hat{O}_1^2$. Proceeding as in the first part of the theorem proves $O_1 \subseteq \hat{O}_1$ on the event $\mathcal{A}$. □

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